



INFORMS Journal on Computing

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To cite this article:

Halil Şen, Kerem Bülbül (2015) A Strong Preemptive Relaxation for Weighted Tardiness and Earliness/Tardiness Problems on Unrelated Parallel Machines. INFORMS Journal on Computing 27(1):135-150. <http://dx.doi.org/10.1287/ijoc.2014.0615>

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A Strong Preemptive Relaxation for Weighted Tardiness and Earliness/Tardiness Problems on Unrelated Parallel Machines

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Research on due date-oriented objectives in the parallel machine environment is at best scarce compared to objectives such as minimizing the makespan or the completion time-related performance measures. Moreover, almost all existing work in this area is focused on the identical parallel machine environment. In this study, we leverage on our previous work on the single machine total weighted tardiness (TWT) and total weighted earliness/tardiness (TWET) problems and develop a new preemptive relaxation for both problems on a bank of unrelated parallel machines. The key contribution of this paper is devising a computationally effective Benders decomposition algorithm to solve the preemptive relaxation formulated as a mixed-integer linear program. The optimal solution of the preemptive relaxation provides a tight lower bound. Moreover, it offers a near-optimal partition of the jobs to the machines. We then exploit recent advances in solving the nonpreemptive single-machine TWT and TWET problems for constructing nonpreemptive solutions of high quality to the original problem. We demonstrate the effectiveness of our approach with instances of up to five machines and 200 jobs.

Keywords: unrelated parallel machines; weighted tardiness; weighted earliness and tardiness; preemptive relaxation; Benders decomposition; transportation problem; lower bound; heuristic

History: Accepted by Karen Aardal, Area Editor for Design and Analysis of Algorithms; received May 2013; revised February 2014; accepted June 2014.

1. Introduction

The prevalence of actual manufacturing environments, where a set of tasks has to be executed on a set of alternate resources, attests to the practical relevance of the parallel machine scheduling environment. For instance, many production steps in semiconductor manufacturing feature unrelated parallel machines because existing machines are augmented over time with machines of newer technology to ramp up production (Shim and Kim 2007a). Another setting in the inspection operations in semiconductor manufacturing creates the context for a recent work by Detienne et al. (2011) on unrelated parallel machines with stepwise individual job cost functions. Several other industries, such as the beverage, printing, and pharmaceutical industries, require processing steps performed by a set of parallel machines (Biskup et al. 2008). Therefore, a thorough understanding of the trade-offs that govern the parallel machine environment is fundamental for the successful operation in many different manufacturing settings.

The scheduling literature is often criticized for its emphasis on the single-machine environment, which is arguably not encountered frequently in today's complex shop floors. However, virtually every scheduling algorithm conceived for multistage production systems

either generalizes or depends on the fundamental principles derived from basic single-machine scheduling problems. A similar argument is valid for the parallel machine environment as well. Decomposition algorithms devised for multistage systems, such as Lagrangian relaxation, Dantzig-Wolfe reformulation, Benders decomposition, and the shifting bottleneck heuristic, give rise to either single- or parallel machine scheduling subproblems that have to be solved many times in an iterative framework. The ultimate performance of such decomposition approaches depends critically on our ability to solve these subproblems with a high solution quality in short computational times. Moreover, from a theoretical perspective the study of parallel machines is the immediate logical extension of single-machine scheduling. For a given partition of the set of jobs over the set of machines, a parallel machine scheduling problem is just a collection of independent single-machine scheduling problems. Therefore, parallel machine scheduling problems are generally regarded as set-partitioning problems for which the complexity of calculating the cost of a partition depends on the difficulty of the underlying single-machine scheduling problem. Motivated by these practical and theoretical considerations, our primary objective in this paper is to

devise a fast and effective mathematical programming-based heuristic for two fundamental due date-related objectives on unrelated parallel machines.

Most of the studies in the scheduling literature are concerned with developing algorithms for a single objective function. These approaches tend to be highly specialized and not easily extended to other objectives and settings. Ultimately, scheduling software is tailored to individual settings, and scheduling research is fragmented. In this context, we emphasize that in this paper we attack two popular scheduling objectives—total weighted tardiness (TWT) and total weighted earliness/tardiness (TWET)—within a single algorithmic framework. The TWT objective is a special case of the TWET objective; however, TWET is nonregular and TWT is regular. It is well established that nonregular objectives give rise to new theoretical and computational issues (Baker and Scudder 1990, Kanet and Sridharan 2000), and it is uncommon to tackle both objectives simultaneously. Formally, we characterize the problems we consider as $Rm // \sum_j \pi_j T_j$ (Rm-TWT) and $Rm // \sum_j \pi_j T_j + \epsilon_j E_j$ (Rm-TWET) for minimizing the TWT and TWET, respectively, on a set of m unrelated parallel machines, following the three-field notation of Graham et al. (1979) in classifying scheduling problems. The notation Rm in the first field stands for a bank of m unrelated machines. The earliness and tardiness of job j are represented by E_j and T_j , respectively, and ϵ_j and π_j are the associated unit weights. Both Rm-TWT and Rm-TWET are strongly $\mathcal{NP}$ -hard because the strongly $\mathcal{NP}$ -hard single-machine scheduling problem $1 // \sum_j \pi_j T_j$ (Lenstra et al. 1977) is a special case of both of these problems. Next we briefly summarize our motivation and the main contributions of this paper.

The review of the related literature in §2 identifies the lack of strong lower bounds as a major impediment to the development of exact algorithms and the performance analysis of heuristics for the TWT and TWET objectives in the parallel machine environment. Shim and Kim (2007a) attack the unweighted version of Rm-TWT, and their branch-and-bound (B&B) algorithm does not scale beyond five machines and 20 jobs. In their concluding remarks, the authors state that “...further research is needed if one needs to solve problems of larger or practical sizes. One way may be to develop more effective or tighter lower bounds since the lower bound used in the B&B algorithm suggested in this study does not seem to be very tight” (p. 354). More generally, in their effort to compute strong linear programming (LP)-based bounds for a class of parallel machine scheduling problems with additive objectives, van den Akker et al. (1999, p. 862) observe that “additive objective functions pose a computational challenge because it is difficult to compute strong lower bounds.” These comments provide a strong motivation for this study. All promising existing results assume that the

machines are identical and often exploit this fact in some way, e.g., by aggregating the machine capacity constraints. Clearly, such approaches do not necessarily extend to or yield similar results for unrelated parallel machines.

In this paper, we set out to provide tight lower bounds and near-optimal solutions for the TWT and TWET objectives in the unrelated parallel machine environment. To this end, we propose a new preemptive relaxation that explicitly assigns jobs to specific machines. This preemptive relaxation generalizes and builds on the success of related previous studies on the single-machine weighted tardiness and weighted earliness/tardiness scheduling problems (Sourd and Kedad-Sidhoum 2003, Bülbül et al. 2007, Pan and Shi 2007, Şen and Bülbül 2012). The resulting lower bound is tight, and perhaps more importantly, the job partition retrieved from the (near-) optimal solution of the preemptive relaxation provides sufficient information to construct feasible nonpreemptive schedules of high quality for the original problem. That is, we recognize that the main practical difficulty of solving Rm-TWT and Rm-TWET to (near-) optimality is determining a good job partition, and we directly incorporate this aspect of the problem into our rationale for developing this particular relaxation. Once a job partition is available, we rely on recent advances by Tanaka et al. (2009) and Tanaka and Fujikuma (2012) to solve m independent single-machine TWT or TWET problems, respectively, to construct a nonpreemptive solution of high quality to the original unrelated parallel machine scheduling problem. The downside of our preemptive relaxation is that it is formulated as a difficult mixed-integer linear program. A key contribution of this paper is devising a computationally effective Benders decomposition algorithm that can handle very large instances of this formulation. Here, the lazy constraint generation scheme of IBM ILOG CPLEX (2011) proves instrumental for a successful implementation. Moreover, as previously shown, both objectives TWT and TWET are tackled successfully by the same algorithm.

In the next section, we review the related literature and put our work into perspective. We introduce and formulate the proposed preemptive relaxation in §3 and then develop our solution approach based on Benders decomposition in §4. This is followed in §5 by an extensive set of computational experiments. We conclude and discuss potential future research directions in §6.

2. Review of Related Literature

Early research on parallel machine scheduling is primarily concerned with the makespan and total (weighted) completion time objectives (Cheng and Sin 1990). We refer the reader to Pinedo (2008) for a comprehensive discussion of the polynomially solvable cases

and structural results of interest for these problems. Some of the more recent and well-known examples of the papers that study $\mathcal{NP}$ -complete problems in this domain include van den Akker et al. (1999), Chen and Powell (1999b), and Azizoglu and Kirca (1999a, b). Studies on due date-related performance measures in the parallel machine environment commenced in earnest in the 1990s and picked up more significantly during the last decade. In this review, we mainly restrict our attention to the literature on parallel machine tardiness and earliness/tardiness scheduling problems with job-dependent due dates. This part of the literature creates the context for our study, and we provide a few important pointers otherwise. The great majority of the existing studies on due date-related performance measures assumes that the machines are identical, and only a handful of papers consider the case of unrelated parallel machines. For most of the proposed exact approaches, computational scalability remains an issue because of the lack of strong lower bounds. Therefore, we also specifically elaborate on the existing lower bounding methods for parallel machine scheduling problems with additive tardiness and earliness/tardiness objective functions to justify our alternate lower bounding scheme, introduced in §3. See Table 1 in the online supplement (available as supplemental material at <http://dx.doi.org/10.1287/ijoc.2014.0615>) for a summary of the important points in this section.

The first exact approach for minimizing the total tardiness with distinct due dates on identical parallel machines is from Azizoglu and Kirca (1998). The authors integrate some dominance rules and a simple bounding technique into a B&B procedure for the problem $Pm//\sum_j T_j$, where Pm in the first field indicates a set of m identical parallel machines. The algorithm is able to handle instances with up to 15 jobs and three machines. The lower bound of Azizoglu and Kirca (1998) belongs to a very common and simple set of lower bounds that rely on determining a lower bound for the j th smallest job completion time $C_{[j]}$, $j = 1, \dots, n$ among the set of all feasible schedules. These lower bounds on the completion times are then matched with the weights and the due dates in some appropriate order so that the resulting expression yields a lower bound for the problem under consideration. Lower bounding techniques based on such *minimal completion times* are developed or employed in several other papers with tardiness-related objectives (see Koulamas 1997; Yalaoui and Chu 2002; Liaw et al. 2003; Shim and Kim 2007a, b; Souayah et al. 2009). There is a consensus in the literature that this class of lower bounds is not strong in general. Furthermore, in problems with earliness/tardiness objectives the presence of unforced idle time renders similar lower bounding techniques invalid.

For the same problem $Pm//\sum_j T_j$, Yalaoui and Chu (2002) devise another B&B scheme. The limit of this algorithm appears to be 20 jobs and two machines within a time limit of 30 minutes. The series of papers by Liaw et al. (2003) and Shim and Kim (2007a, b) develop a set of closely related optimal methods. Liaw et al. (2003) attack the problem $Rm//\sum_j \pi_j T_j$ of minimizing TWT on unrelated parallel machines. This study appears to be the first exact approach for this problem. The lower bounding scheme is very similar to that in Azizoglu and Kirca (1999b), with a simple enhancement based on the structure of the tardiness objective; however, the method does not scale beyond four machines and 18 jobs. Shim and Kim (2007a) tackle the unweighted version $Rm//\sum_j T_j$ in the same machine environment. The proposed B&B method employs some of the existing dominance properties in addition to new ones. The lower bounding technique from Liaw et al. (2003) is adopted, and an alternate lower bound is obtained by reducing the original problem into a single-machine problem by modifying the processing times appropriately and using a previously existing result for the single-machine total tardiness problem. The largest problem size that can be handled successfully within one hour is five machines and 20 jobs. In a similar work, Shim and Kim (2007b) address the problem $Pm//\sum T_j$, and instances with up to five machines and 30 jobs are solved optimally within one hour. Jouglet and Savourey (2011) devise dominance rules and filtering methods for the problem $Pm/r_j/\sum_j \pi_j T_j$, where the notation r_j in the second field indicates that the release dates may be nonidentical, and embed these into a B&B procedure along with an existing lower bound. The authors argue that the lack of good lower bounds prevents them from solving instances with more than 20 jobs and three machines.

All optimal methods discussed so far base their lower bounding efforts on combinatorial arguments that rely on simple properties of the scheduling objectives under consideration. The resulting bounds are generally loose. However, the most promising lower bounds for parallel machine total (weighted) tardiness and earliness/tardiness problems are derived through mathematical programming techniques. For instance, the LP relaxations of the set partitioning formulations of common due date/common due window earliness/tardiness problems solved by column generation yield a prominent class of tight lower bounds (Chen and Powell 1999a, Chen and Lee 2002). **Bounds obtained from various relaxations of time-indexed formulations are also popular in parallel machine scheduling. An arc-time-indexed formulation whose LP relaxation is tackled by column generation is at the heart of the highly efficient branch-cut-and-price algorithm of Pessoa et al. (2010) for $Pm//\sum \pi_j T_j$. This study is by far the most successful exact algorithm**

to date on parallel machine tardiness problems and delivers optimal solutions for instances with up to 100 jobs and four machines. Tanaka and Araki (2008) apply Lagrangian relaxation to the time-indexed formulation of the problem $Pm//\sum T_j$ in an effort to develop tight lower bounds. Instances with up to 25 jobs and 10 machines are solved optimally. The average gap of the initial lower bound is 2.4% for the instances not solved at the root node. Souayah et al. (2009) take on the weighted version of the problem and study $Pm//\sum_j \pi_j T_j$. With a mix of combinatorial, mathematical programming and Lagrangian relaxation-based lower bounds, about half of the instances with up to 35 jobs and two machines are solved to optimality within 20 minutes. We refer the interested reader to the review paper Sen et al. (2003), where the tardiness literature on multimachine systems is briefly addressed as well. Following this discussion are two observations regarding the state of the literature. First, there is a clear need to study the tardiness-related objectives in the unrelated parallel machine environment; we can pinpoint only two studies that focus on the unrelated parallel machine environment. Second, more than 20 jobs and a few machines seems to be generally beyond the reach of the existing exact methods, which is attributed to the lack of strong lower bounds. We hope to provide a potential remedy to this issue.

Several heuristics have been proposed for minimizing the total (weighted) tardiness on identical parallel machines; these are reviewed in §1.1 in the online supplement. For unrelated parallel machines, we are aware of only three papers—Zhou et al. (2007), Mönch (2008), and Lin et al. (2011)—that focus on heuristics for $Rm//\sum_j \pi_j T_j$. The first two studies rely on ant colony optimization and benchmark their algorithms against simple heuristics, which makes it difficult to evaluate the solution quality in absolute terms. Lin et al. (2011) propose a genetic algorithm and two simpler heuristics. The genetic algorithm outperforms all others in the computational experiments and deviates from the optimal solution by 1.8% on average for small instances with four machines and 20 jobs. The heuristic that we develop in this paper is scalable to large instances with up to 200 jobs; it simultaneously produces both lower and upper bounds of high quality. As evident from the discussion, this is a significant edge over other studies in the literature, and we make a valuable contribution to the (unrelated) parallel machine scheduling research with tardiness objectives.

To the best of our knowledge, no exact algorithm has been designed to date for the problem of scheduling a set of independent jobs on a bank of unrelated parallel machines with the objective of minimizing the total (weighted) earliness and tardiness. However, various studies investigate special cases of this problem—see §1.2 in the online supplement. The most

closely related works to our problem Rm -TWET are by Kedad-Sidhoum et al. (2008), Mason et al. (2009), and M'Hallah and Al-Khamis (2012). Kedad-Sidhoum et al. (2008) experiment with various relaxations of the problem $Pm/r_j/\sum_j \pi_j T_j + \epsilon_j E_j$ by recognizing that the main difficulty in solving earliness/tardiness scheduling problems stems from the lack of strong lower bounds. The authors extend two classes of lower bounds originally proposed for the single-machine case to the identical parallel machine environment. Their discrete assignment-based lower bound is discussed further in §3 because it is closely related to our preemptive lower bounding method for Rm -TWT and Rm -TWET. That study reports that the Lagrangian relaxation obtained by dualizing the machine capacity constraints in the time-indexed formulation outperforms others, taking into account both the solution quality and gap. Tanaka and Araki (2008)—discussed previously—employ the same Lagrangian relaxation for $Pm//\sum T_j$. The best bound attained by solving the Lagrangian dual problem in these relaxations is equivalent to that provided by the LP relaxation of the time-indexed formulation. However, solving the Lagrangian dual problem—generally by subgradient optimization—is often computationally more efficient. We also attest to the rapidly increasing computational effort required to solve the LP relaxation of the time-indexed formulation in §5. Kedad-Sidhoum et al. (2008) obtain upper bounds through a simple local search. Experimental results attest to the quality of both the lower and upper bounds. The average optimality gap attained for instances with up to six machines and 90 jobs is around 1.5%.

However, we cite two good reasons for not following a similar path to that of Kedad-Sidhoum et al. (2008) and Tanaka and Araki (2008). First, the machine capacity constraints in the time-indexed formulation may be aggregated in the identical parallel machine environment by defining a single resource with a capacity of performing m jobs simultaneously, and this renders the number of dual variables in the Lagrangian relaxation independent from the number of machines in the problem. This, however, is not possible for Rm -TWT and Rm -TWET, and relaxing the machine capacity constraints—one for each combination of time period and machine—would result in mH dual variables instead of just H . Consequently, solving the Lagrangian dual problem would quickly become a formidable task with an increasing number of machines. Second, the solution retrieved from the Lagrangian relaxation offers little information on how to identify near-optimal job to machine assignments. The job start times provided by the Lagrangian relaxation for a given set of dual multipliers form the basis for a dispatch rule in Tanaka and Araki (2008); however, both these authors and Kedad-Sidhoum et al. (2008) need to devise independent heuristics to obtain high-quality feasible solutions for their original problems.

The moving block heuristic of Mason et al. (2009) for $Pm//\sum E_j + T_j$ is tested against an integer programming formulation over instances with up to 40 jobs and four machines. The heuristic identifies feasible solutions that are on average better than the incumbent for 20- and 40-job instances. Like Kedad-Sidhoum et al. (2008), M'Hallah and Al-Khamis (2012) tackle the weighted version of the problem. Their integer programming formulation points out and corrects an error in the study by Mason et al. (2009). The limit of the formulation appears to be instances with no more than 20 jobs. In addition, several new heuristics are introduced. The best performing contender turns out to be a hybrid heuristic that is benchmarked against the lower and upper bounds of Kedad-Sidhoum et al. (2008). The hybrid heuristic improves some of the best known solutions for the instances of Kedad-Sidhoum et al. (2008); however, it yields slightly worse solutions on average. The median gap of the hybrid heuristic ranges from 1.4% to 6.1% with respect to the lower bounds of Kedad-Sidhoum et al. (2008), depending on the problem size. It is evident that there is a gap in the literature with respect to the parallel machine earliness/tardiness scheduling problems with distinct due dates. To the best of our knowledge, our work provides the first viable solution approach for the unrelated parallel machine environment in this context.

3. Problem Statement and Preemptive Relaxation

We consider a bank of m unrelated parallel machines and n jobs, which are all ready at time zero. Each job is processed on exactly one of the machines, where the processing of job j on machine k requires an integer duration of p_{jk} time units. The completion time of job j is denoted by C_j . A due date d_j —also assumed to be integral—is associated with each job j , and we incur a cost π_j per unit time if job j completes processing after d_j . Thus, the total weighted tardiness over all jobs is determined as $\sum_j \pi_j T_j$, where the tardiness of job j is calculated as $T_j = \max(0, C_j - d_j)$. For the problem $Rm//\sum_j \pi_j T_j + \epsilon_j E_j$, the objective additionally penalizes the completion of job j prior to its due date d_j at a rate of ϵ_j per unit time, where the earliness of job j is defined as $E_j = \max(0, d_j - C_j)$. All machines are available continuously from time zero onward, and a machine can execute at most one operation at a time. An operation must be carried out to completion once started; i.e., preemption is not allowed.

In this section, we introduce our preemptive lower bounding scheme for Rm-TWT and Rm-TWET. We define two primary design goals for our preemptive relaxation. The tightness of the lower bound is clearly a major concern. Equally important is the information that can be extracted from the optimal solution of the

preemptive relaxation to construct feasible solutions of high quality for the original nonpreemptive problem. We attain both goals—somewhat more successfully for Rm-TWT than for Rm-TWET from a computational perspective—and demonstrate the effectiveness of the proposed lower and upper bounds in §5.

A class of highly efficient lower bounds based on a particular preemption scheme was developed for single-machine tardiness and earliness/tardiness scheduling problems during the last decade (Bülbül et al. 2007, Sourd and Kedad-Sidhoum 2003, Şen and Bülbül 2012). The key idea of these preemptive relaxations is to divide up jobs with integer processing times into jobs of unit length and associate a cost with the completion of each unit length job. That is, jobs may only be preempted at integer points in time. In this setting, the problem of solving the preemptive relaxation is formulated as an assignment or a transportation problem, where the length of the planning horizon depends on the magnitude of the due dates and the sum of the processing times. Therefore, the formulation size is pseudopolynomial. On the positive side, the availability of very fast algorithms for the assignment and transportation problems still renders this lower bounding technique viable. The following formulation (TR) is from Kedad-Sidhoum et al. (2008), where the original approach in the single-machine environment is extended to m identical parallel machines:

$$(TR) \text{ minimize } \sum_{j=1}^n \sum_{t=1}^H c'_{jt} x_{jt} \quad (1)$$

$$\text{subject to } \sum_{t=1}^H x_{jt} = p_j, \quad j=1, \dots, n, \quad (2)$$

$$\sum_{j=1}^n x_{jt} \leq m, \quad t=1, \dots, H, \quad (3)$$

$$0 \leq x_{jt} \leq 1, \quad j=1, \dots, n, t=1, \dots, H. \quad (4)$$

In the model (TR), the time period t represents the time interval $(t-1, t]$, and consequently in any optimal schedule all jobs finish processing no later than in period H , where

$$H = \begin{cases} \left\lceil \frac{\sum_{j=1}^n \max_k(p_{jk})}{m} \right\rceil + p_{\max}, & \text{for Rm-TWT;} \\ \left\lceil \frac{\sum_{j=1}^n \max_k(p_{jk})}{m} \right\rceil + p_{\max} + d_{\max}, & \text{for Rm-TWET.} \end{cases} \quad (5)$$

The end of the planning horizon H is determined based on the following observation: For Rm-TWT with a regular objective function, all machines are continuously

busy until some time $t' \leq \lceil \sum_{j=1}^n \max_k(p_{jk})/m \rceil$ if at least m jobs are still not completed. Therefore, after time t' the remaining $m-1$ jobs are finished in at most $p_{\max} = \max_{j,k}(p_{jk})$ time periods. The end of the planning horizon may thus be set to the value in the first row of (5). An optimal solution of Rm-TWET, in contrast, may include unforced idleness, and the argument just described is only valid if we conservatively assume that all jobs start at $d_{\max} = \max_j d_j$. Clearly, $p_{\max}$ may be omitted from (5) in the case of a single machine.

If a unit job of job j is executed during the time interval $(t-1, t]$, the decision variable x_{jt} assumes the value 1, and the objective is charged a cost of c'_{jt} . The constraints (2) mandate that each job j receives p_j units of processing. To observe the machine capacities, constraints (3) require that no more than m unit jobs are processed simultaneously in a given period. Note that the machine index is omitted from the processing times because they are all identical for a given job. Furthermore, no integrality is imposed on the decision variables because of the total unimodularity of the constraint matrix of (TR). The optimal objective function value of (TR) is a lower bound on that of $Pm // \sum_j \pi_j T_j + \epsilon_j E_j$, as long as the objective function coefficients satisfy

$$\sum_{s=t-p_j+1}^t c'_{js} \leq \epsilon_j (d_j - t)^+ + \pi_j (t - d_j)^+, \quad j = 1, \dots, n, \quad t = p_j, \dots, H. \quad (6)$$

That is, the total cost incurred in (TR) by any job that is scheduled nonpreemptively is no larger than that of the original nonpreemptive problem (Bülbül et al. 2007). Naturally, the strength of the lower bound depends on the objective coefficients c'_{jt} , and this is where the existing works in the literature take different paths. For instance, the cost coefficients of Sourd and Kedad-Sidhoum (2003) satisfy (6) as an equality. Bülbül et al. (2007) characterize and develop an expression for the cost coefficients that are the best among those with a piecewise linear structure with two segments. For these cost coefficients, (6) holds as a strict inequality for some values of t . For the single-machine problem, these authors also show that the lower bound retrieved from (TR) is no better than that provided by the LP relaxation of the time-indexed formulation. Conversely, Pan and Shi (2007) prove the existence of a set of objective coefficients for (TR) so that the LP relaxation of the time-indexed formulation and (TR) yield identical lower bounds. However, computing the values of these cost coefficients is no less time consuming than solving the LP relaxation of the time-indexed formulation. We also note that the empirical performance of the algorithms based on this set of relaxations is more than satisfactory (Sourd and Kedad-Sidhoum 2003, Bülbül et al. 2007, Pan and Shi 2007, Şen and Bülbül 2012).

They strike a good balance between solution quality and time.

Factoring in all arguments in this section, the set of preemptive relaxations discussed in the previous paragraph emerges as a strong candidate for deriving strong lower bounds for our problems of interest Rm-TWT and Rm-TWET. However, one hurdle remains in the pursuit of our second design goal of constructing nonpreemptive solutions of high quality directly based on the information retrieved from the optimal solution of the preemptive relaxation. In the optimal solution of (TR), the unit jobs of job j cannot overlap in time, but they can be processed on different machines. Consequently, no explicit assignment of the jobs to the machines is available. This is a major drawback because it complicates the task of obtaining a nonpreemptive feasible solution to the original problem. In the following, we demonstrate that overcoming this difficulty allows us to attain good upper bounds in addition to good lower bounds.

The downside of (TR) is that the optimal solution does not guarantee that we can assign all unit jobs of a job to the same machine. Such a requirement is incorporated in the following model (TR-A) at the expense of additional variables and destroying the desirable polyhedral structure of the transportation problem. The binary variable y_{jk} takes the value 1 if job j is assigned to machine k , and is 0 otherwise. In addition, the x -variables and the associated objective coefficients are supplemented with a machine index k to allow us to assign a unit job of job j explicitly to machine k in period t .

$$(TR-A) \quad \text{minimize} \quad \sum_{j=1}^n \sum_{k=1}^m \sum_{t=1}^H c'_{jkt} x_{jkt} \quad (7)$$

$$\text{subject to} \quad \sum_{t=1}^H x_{jkt} = p_j y_{jk}, \quad j = 1, \dots, n, \quad k = 1, \dots, m, \quad (8)$$

$$\sum_{j=1}^n x_{jkt} \leq 1, \quad k = 1, \dots, m, \quad t = 1, \dots, H, \quad (9)$$

$$\sum_{k=1}^m y_{jk} = 1, \quad j = 1, \dots, n, \quad (10)$$

$$x_{jkt} \geq 0, \quad j = 1, \dots, n, \quad k = 1, \dots, m, \quad t = 1, \dots, H, \quad (11)$$

$$y_{jk} \in \{0, 1\}, \quad j = 1, \dots, n, \quad k = 1, \dots, m. \quad (12)$$

(TR-A) differs from (TR) in two main aspects. The capacity constraints (9) appear in a disaggregated form, and all unit jobs of job j are performed on the same

machine by constraints (8) and the job-partitioning constraints (10). As we hinted at earlier, the cost coefficients c'_{jkt} are of critical importance for the strength of the lower bounds provided by the preemptive relaxation. In this research, we stick with the cost coefficients by Bülbül et al. (2007) given in (13) and adapted in an obvious way to the unrelated parallel machine environment for two reasons: they empirically outperform those by Sourd and Kedad-Sidhoum (2003) on average (Bülbül et al. 2007, Kedad-Sidhoum et al. 2008), and computing the best set of cost coefficients for a given instance by the method of Pan and Shi (2007) is expensive.

$$c'_{jkt} = \begin{cases} \frac{\epsilon_j}{p_{jk}} \left[\left(d_j - \frac{p_{jk}}{2} \right) - \left(t - \frac{1}{2} \right) \right], & \text{for } t \leq d_j; \\ \frac{\pi_j}{p_{jk}} \left[\left(t - \frac{1}{2} \right) - \left(d_j - \frac{p_{jk}}{2} \right) \right], & \text{for } t > d_j. \end{cases} \quad (13)$$

We next provide a proposition that the optimal solution of (TR-A) with the cost coefficients given earlier provides a lower bound on the optimal objective function value of the original problem Rm-TWT or Rm-TWET. The result is a corollary of (Bülbül et al. 2007, Theorem 2.2), where the authors show that the cost coefficients in (13) satisfy (6). The formal proof is in §2.1 of the online supplement.

PROPOSITION 1. *The optimal objective function value of (TR-A) with the cost coefficients given by equation (13) is a lower bound on the optimal objective function value of the original nonpreemptive problem Rm-TWT or Rm-TWET.*

Our overall strategy for obtaining near-optimal feasible solutions and good lower bounds for Rm-TWT and Rm-TWET is now clear. We first solve (TR-A), retrieve the job partition, and then build m individual machine schedules independently. Several heuristics with excellent empirical performance are available for both $1/\sum_j \pi_j T_j$ and $1/\sum_j \pi_j T_j + \epsilon_j E_j$ to perform the latter task. However, in this work we rely on the powerful optimal algorithms of Tanaka et al. (2009) and Tanaka and Fujikuma (2012) to handle the single-machine problems, as mentioned in §1. Our computational experiments in §5 ultimately support this decision. Thus, only one major challenge remains. The formulation (TR-A) is a mixed-integer programming problem that is time consuming to solve based on our preliminary computational experiments. However, for a fixed job partition it decomposes into m independent LPs— m independent transportation problems—and these LPs are solved to optimality very efficiently. These observations suggest that (TR-A) is amenable to Benders decomposition (Benders 1962), and developing a Benders decomposition algorithm with *strengthened cuts* for (TR-A) is our main methodological contribution.

One final remark before we delve into the specifics of our solution method for (TR-A): For Rm-TWT, the formulation may be strengthened by the load-balancing constraints (14), which assert that the workloads of two machines cannot differ by more than $p_{\max}$ in an optimal solution of the original nonpreemptive parallel machine scheduling problem. Otherwise, we could transfer the final job on one of these machines to the other one without degrading the objective function value. Note that similar concepts have been incorporated into various properties and dominance rules elsewhere in the literature (Azizoglu and Kirca 1999b, Theorem 1). However, for Rm-TWET with a nonregular objective function, we can easily create instances for which no optimal solution satisfies (14):

$$-p_{\max} \leq \sum_{j=1}^n p_{jk} y_{jk} - \sum_{j=1}^n p_{jl} y_{jl} \leq p_{\max}, \quad k = 1, \dots, m-1, l = k+1, \dots, m. \quad (14)$$

These cuts are added to the preemptive formulation (TR-A) when solving Rm-TWT and help speed the solution process for large instances.

4. Benders Decomposition

Parallel machine scheduling problems have both a partitioning and a scheduling component. That is, if we assign jobs to machines by fixing the variables y_{jk} , $j = 1, \dots, n$, $k = 1, \dots, m$, so that the constraints (10) are satisfied, then the model (TR-A) decomposes into m independent transportation problems. We exploit this key observation to design an algorithm based on Benders decomposition for solving (TR-A) efficiently. To this end, we reformulate (TR-A) for a fixed $\bar{y}$ by replacing the right-hand side of the set of constraints (8) by $p_{jk} \bar{y}_{jk}$ and dropping the set of constraints (10) and (12) from the model. In the resulting linear program (TR-A($\bar{y}$)), u_{jk} , $j = 1, \dots, n$, $k = 1, \dots, m$, and v_{kt} , $k = 1, \dots, m$, $t = 1, \dots, H$, are the dual variables associated with the set of constraints (8) and (9), respectively. The dual of (TR-A($\bar{y}$)) is stated below, where the decomposition into m independent transportation problems is made explicit:

$$z(\bar{y}) = \sum_{k=1}^m z_k(\bar{y}), \quad (15)$$

where

$$(DS_k) \quad z_k(\bar{y}) = \text{maximize} \quad \sum_{j=1}^n p_{jk} \bar{y}_{jk} u_{jk} + \sum_{t=1}^H v_{kt} \quad (16)$$

$$\text{subject to} \quad u_{jk} + v_{kt} \leq c'_{jkt}, \quad j = 1, \dots, n, t = 1, \dots, H, \quad (17)$$

$$v_{kt} \leq 0, \quad t = 1, \dots, H, \quad (18)$$

is the dual of the transportation problem (TR_k) for machine k . In the following, (TR_k) and (DS_k) are also referred to as the *cut generation subproblem* and the *dual slave problem*, respectively, by following the common terminology for Benders decomposition.

Based on the objective function (16) of (DS_k), we obtain the following relaxed Benders master problem (RMP), where C denotes the current number of times the cut generation subproblems (TR_k), $k = 1, \dots, m$, have been solved. The optimal values of the dual variables $\bar{u}_{jk}$, $j = 1, \dots, n$, $k = 1, \dots, m$, and $\bar{v}_{kt}$, $k = 1, \dots, m$, $t = 1, \dots, H$, in round c of the cut generation are represented by $\bar{u}_{jk}^c$ and $\bar{v}_{kt}^c$, respectively. The auxiliary variable η_k indicates a lower bound on the total cost incurred by the jobs assigned to machine k , and the objective function value $\sum_{k=1}^m \eta_k$ of (RMP) is therefore a lower bound on the optimal objective values of (TR-A) and the original nonpreemptive scheduling problem Rm-TWT or Rm-TWET.

$$\text{(RMP) minimize } \sum_{k=1}^m \eta_k \quad (19)$$

$$\text{subject to } \sum_{k=1}^m y_{jk} = 1, \quad j = 1, \dots, n, \quad (20)$$

$$\eta_k \geq \sum_{j=1}^n p_{jk} \bar{u}_{jk}^c y_{jk} + \sum_{t=1}^H \bar{v}_{kt}^c, \quad k = 1, \dots, m, c = 1, \dots, C, \quad (21)$$

$$y_{jk} \in \{0, 1\}, \quad j = 1, \dots, n, k = 1, \dots, m. \quad (22)$$

Note that (TR-A($\bar{y}$)) is feasible and (DS_k), $k = 1, \dots, m$, is bounded for any $\bar{y}$ that satisfies the constraints (10). Therefore, no feasibility cuts are required, and only optimality cuts are generated and added iteratively to (RMP) during the course of the algorithm. Furthermore, the cut generation subproblem (TR_k) for machine k includes all jobs and is solved by considering the full length of the planning horizon. From a computational point of view, however, we are better off defining the set of jobs to be processed on machine k as $J_k = \{j \mid y_{jk} = 1\}$, setting the last period of processing on machine k —designated by H_k —as appropriate based on (5), and then solving a restricted version of (TR_k) over these jobs and time periods only. This restricted cut generation subproblem formulation and the corresponding dual slave problem are referred to as (TR_k-R) and (DS_k-R), respectively. Obviously, the optimal solution of (TR_k-R) may be extended to an optimal solution of (TR_k) trivially by setting $x_{jkt} = 0$ for $j \in J_k$, $t = H_k + 1, \dots, H$, and $j \notin J_k$, $t = 1, \dots, H$. The relationship between the optimal solutions of (DS_k) and (DS_k-R) and the implications for the dynamic generation of the constraints (21) require deeper discussion; see §4.1.

In the *multicut* relaxed master problem formulation (RMP), we approximate the objective function of (TR-A) by estimating the cost accumulated on each machine separately, as evident from the set of constraints (21). Alternatively, we could have employed a weaker *single-cut* version of the relaxed master problem by aggregating all m cuts generated after solving (TR_k), $k = 1, \dots, m$, and replacing $\sum_{k=1}^m \eta_k$ with a single variable η in the formulation, as appropriate. The single-cut version results in fast solution times for the relaxed master problem at the expense of more iterations overall. Ultimately, the trade-off between these two alternatives is only decided during the computations. In our preliminary testing, the cut generation algorithm based on (RMP) was clearly superior to that based on the single-cut version in terms of speed. Thus, the rest of the paper is focused exclusively on (RMP). The pseudocode of the cut generation procedure is stated in Algorithm 1 in §3 of the online supplement.

4.1. Validity and Strengthening of the Benders Cuts

The validity of Benders decomposition (Benders 1962) derives from the independence of the feasible region of the dual slave problem from the values of the integer variables. For a mixed-integer programming problem of the form minimize $\{ \mathbf{g}\mathbf{x} + \mathbf{h}\mathbf{y} : \mathbf{G}\mathbf{x} + \mathbf{H}\mathbf{y} \geq \mathbf{b}, \mathbf{x} \in \mathbb{R}^+, \mathbf{y} \in \mathbb{Z}^+ \}$, where all matrices and vectors have appropriate dimensions, the dual slave problem for a given $\bar{\mathbf{y}}$ is stated as maximize $\{ \mathbf{w}^T(\mathbf{b} - \mathbf{H}\bar{\mathbf{y}}) : \mathbf{w}^T \mathbf{G} \leq \mathbf{g}, \mathbf{w} \in \mathbb{R}^+ \}$, where $\mathbf{w}$ is the vector of dual variables of appropriate size. In other words, the dual slave problem is always solved over the same dual polyhedron $\{ \mathbf{w}^T \mathbf{G} \leq \mathbf{g}, \mathbf{w} \in \mathbb{R}^+ \}$, and only the objective function depends on the values of the integer variables. As a consequence, the maximum number of cuts to be generated is bounded from above by the number of extreme points of the dual polyhedron. These issues need a closer look, however, if we opt to solve (TR_k-R) instead of (TR_k) because this amounts to solving the dual slave problem over different feasible regions every time and contradicts the basic pillar of Benders decomposition. Observe that a cut of the form

$$\eta_k \geq \sum_{j \in J_k} p_{jk} \bar{u}_{jk} y_{jk} + \sum_{t=1}^{H_k} \bar{v}_{kt} \quad (23)$$

produced directly out of an optimal solution of (DS_k-R) relies on the assumption that augmenting this solution trivially with $\bar{u}_{jk} = 0$ for $j \notin J_k$ and $\bar{v}_{kt} = 0$ for $t = H_k + 1, \dots, H$, is feasible with respect to (DS_k). It is a simple matter to show that as long as the optimal solution of (DS_k-R) satisfies $\max_{t=1, \dots, H_k} \bar{v}_{kt} = 0$ (see Lemma 1), this augmented solution is feasible with respect to (DS_k) if we are solving an instance of Rm-TWT because the cost coefficients c'_{jkt} are nonnegative and nondecreasing over time. However, the trivial augmentation is not

necessarily feasible for every instance of Rm-TWET, and (23) might therefore be an invalid Benders cut. To illustrate, consider an instance of Rm-TWET and assume that for some assignment $\bar{y}$ of the jobs to the machines we solve (TR_k-R) with $H_k \leq d_j - 1$, where $j \notin J_k$ and $p_{jk} > 1$. In the trivially augmented solution for (DS_k), constraint (17) for job j and time period d_j is violated because $c'_{jkd_j} = (\epsilon_j/p_{jk})(1/2 - p_{jk}/2) < 0$ for $\epsilon_j > 0$ and $p_{jk} > 1$, and $\bar{u}_{jk} + \bar{v}_{kd_j} = 0 + 0 \leq c'_{jkd_j}$ does not hold. Therefore, we need a mechanism that can always extend an optimal solution of (DS_k-R) to an optimal solution of (DS_k). Proposition 2 proves that the cut strengthening procedure described next fulfills this goal. This ensures that the dual slave problem is always solved over the same feasible region and that the generated Benders cuts are valid.

Several papers report that a straightforward implementation of Benders decomposition yields a dismal performance from a computational point of view (Van Roy 1986, Üster and Agrahari 2011, Magnanti and Wong 1981, Wentges 1996, Fischetti et al. 2010). This is often rooted in the primal degeneracy in the cut generation subproblem, which implies the existence of multiple optimal solutions to the dual slave problem. That is, possibly several alternate cuts may be generated based on the same master problem solution, and the particular choice has a profound impact on the computational performance. These concerns are also valid for us because the transportation problem suffers from a well-known primal degeneracy.

To address these issues, we initially adapted the generic Benders cut strengthening method introduced by Fischetti et al. (2010) to our problem. These authors argue that identifying a small set of constraints in the subproblem that allows us to cut the current master solution is of practical interest to enhance the computational performance. To this end, they pose the cut generation subproblem as a pure feasibility problem and look for a *minimal infeasible subsystem* of small cardinality. However, applying this technique to our problem does not preserve the transportation problem structure in the cut generation subproblems. This results in substantially prolonged subproblem solution times and ultimately uncompetitive overall performance for Benders decomposition. Instead, we follow an approach here that is similar to those of Van Roy (1986) and Üster and Agrahari (2011) to strengthen our Benders cuts, which also resolves the issue pointed out above regarding the validity of the cuts constructed based on an optimal solution of (DS_k-R). The result is great savings in solution time from this enhancement. In fact, our algorithm exhibits very poor convergence without this cut strengthening.

The key to showing the validity of our cut generation as well as strengthening the Benders cuts is to prove that we can always augment an optimal solution of

(DS_k-R) to obtain a feasible solution of (DS_k) with the same objective function value. This would establish that the augmented solution is optimal for (DS_k) because $\bar{y}_{jk} = 0$ for all $j \notin J_k$ and $v_{kt} \leq 0$ for all $t = H_k + 1, \dots, H$ (see Proposition 2). Compared to (21), the benefit is that we can produce a strengthened Benders cut of the form

$$\eta_k \geq \sum_{j=1}^n p_{jk} \bar{u}''_{jk} y_{jk} + \sum_{t=1}^H \bar{v}''_{kt} \quad (24)$$

from an optimal solution $(\bar{u}''_k, \bar{v}''_k)$ of (DS_k) so that $\bar{u}''_{jk} \neq 0$ for $j \notin J_k$ in general. We first need the following result to attain our goal. The formal proof is in §2.2 of the online supplement.

LEMMA 1. *There exists an optimal solution $(\bar{u}'_k, \bar{v}'_k)$ to (DS_k-R) such that $\max_{t=1, \dots, H_k} \bar{v}'_{kt} = 0$.*

Assume that we are given an optimal solution $(\bar{u}'_k, \bar{v}'_k)$ of (DS_k-R) that satisfies the property in Lemma 1 and corresponding Benders cut of the form (23). Clearly, we can always extend the planning horizon in (DS_k-R) to $1, \dots, H$, and augment this optimal solution with zeros as necessary and still preserve the optimality. Therefore, without loss of generality we assume that an augmented optimal solution $(\bar{u}''_k, \bar{v}''_k)$ is available to (DS_k-R), where $\bar{v}''_{kt} = \bar{v}'_{kt}$ for $t = 1, \dots, H_k$, and $\bar{v}''_{kt} = 0$ for $t = H_k + 1, \dots, H$. Based on this augmented optimal solution, we next explain how an original Benders cut of the form (23) is strengthened and then prove that this strengthened cut corresponds to an optimal solution of (DS_k) and is therefore valid.

The variables u_{jk} , $j \notin J_k$ do not appear in (DS_k-R) and are implicitly assumed to be 0. Consequently, no term appears on the right-hand side of a Benders cut (23) for the jobs that are assigned to other machines in the current relaxed master solution $\bar{y}$. However, $\bar{y}_{jk} = 0$ for all such jobs $j \notin J_k$, and we can produce a stronger cut by incorporating y_{jk} , $j \notin J_k$ into the right-hand side of (23) with positive coefficients $p_{jk} \bar{u}''_{jk}$, $j \notin J_k$, if possible. To compute a good set of values $\bar{u}''_{jk}$, $j \notin J_k$, we solve the following optimization problem for a given augmented optimal solution $(\bar{u}''_k, \bar{v}''_k)$ of (DS_k-R):

$$\text{Maximize } \sum_{j \notin J_k} p_{jk} u_{jk} \quad (25)$$

$$\text{subject to } u_{jk} \leq c'_{jkt} - \bar{v}''_{kt}, \quad j \notin J_k, t = 1, \dots, H. \quad (26)$$

The constraints (26) are required to establish that the coefficients of the strengthened cut correspond to an optimal solution of (DS_k)—see Proposition 2. Clearly, (25) and (26) decompose by job, and the optimal solution is determined as

$$\bar{u}''_{jk} = \min \left\{ \min_{t=1, \dots, H_k} (c'_{jkt} - \bar{v}''_{kt}), \min_{t=H_k+1, \dots, H} c'_{jkt} \right\}, \quad j \notin J_k. \quad (27)$$

For an instance of Rm-TWT, the cost coefficients c'_{jkt} are nondecreasing over $t = 1, \dots, H$. In addition, we have $\max_{t=1, \dots, H_k} \bar{v}''_{kt} = 0$. Then,

$$\min_{t=1, \dots, H_k} (c'_{jkt} - \bar{v}''_{kt}) \leq \max_{t=1, \dots, H_k} c'_{jkt} \leq \min_{t=H_k+1, \dots, H} c'_{jkt}. \quad (28)$$

Consequently, (27) simplifies to

$$\bar{u}''_{jk} = \min_{t=1, \dots, H_k} (c'_{jkt} - \bar{v}''_{kt}), \quad j \notin J_k \quad (29)$$

for Rm-TWT.

For Rm-TWET, we have to differentiate between two cases because the cost coefficients c'_{jkt} , $1, \dots, H$ are not nondecreasing over time:

$$\bar{u}''_{jk} = \begin{cases} \min \left(\min_{t=1, \dots, H_k} (c'_{jkt} - \bar{v}''_{kt}), c'_{jkH_{k+1}} \right), & \text{if } H_k \geq d_j; \\ \min \left(\min_{t=1, \dots, H_k} (c'_{jkt} - \bar{v}''_{kt}), c'_{jkd_j} \right), & \text{if } H_k \leq d_j - 1, \end{cases}, \quad j \notin J_k. \quad (30)$$

Thus, the strengthened cut finally takes the form specified in (24), where $\bar{u}''_{jk} = \bar{u}''_{jk}$ for $j \in J_k$ and $\bar{u}''_{jk}$, $j \notin J_k$ is calculated based on either (29) or (30), respectively, depending on whether we solve an instance of Rm-TWT or Rm-TWET. We next prove that this augmented solution $(\bar{\mathbf{u}}''_k, \bar{\mathbf{v}}''_k)$ is optimal for (DS_k) .

PROPOSITION 2. *The dual variables $(\bar{\mathbf{u}}''_k, \bar{\mathbf{v}}''_k)$, which produce a strengthened Benders cut (24), are optimal with respect to (DS_k) .*

PROOF. Recall that $(\bar{\mathbf{u}}''_k, \bar{\mathbf{v}}''_k)$ is constructed by augmenting an optimal solution $(\bar{\mathbf{u}}'_k, \bar{\mathbf{v}}'_k)$ of $(DS_k\text{-R})$, which satisfies the property in Lemma 1. Therefore, $\bar{u}''_{jk} + \bar{v}''_{kt} \leq c'_{jkt}$, $j \in J_k$, $t = 1, \dots, H_k$, and $\bar{v}''_{kt} \leq 0$, $t = 1, \dots, H_k$, hold automatically. In addition, $\bar{v}''_{kt}$, $t = H_k + 1, \dots, H$, are set directly to zero. Therefore, we only need to verify that $\bar{u}''_{jk} + \bar{v}''_{kt} \leq c'_{jkt}$, $j \in J_k$, $t = H_k + 1, \dots, H$, and $\bar{u}''_{jk} + \bar{v}''_{kt} \leq c'_{jkt}$, $j \notin J_k$, $t = 1, \dots, H$ to show the feasibility of $(\bar{\mathbf{u}}''_k, \bar{\mathbf{v}}''_k)$ for (DS_k) . The latter inequalities are enforced directly by the constraints (26). For the former, note that for any job $j \in J_k$ the end of the planning horizon H_k is larger than d_j in both Rm-TWT and Rm-TWET. Then, by a similar argument that leads to (28), $\bar{u}''_{jk} \leq \max_{t'=1, \dots, H_k} c'_{jkt'}$, and we obtain $\bar{u}''_{jk} + \bar{v}''_{kt} = \bar{u}''_{jk} \leq \max_{t'=1, \dots, H_k} c'_{jkt'} \leq c'_{jkt}$ for all time periods $t = H_k + 1, \dots, H$, as desired.

The optimal objective function value of (DS_k) is bounded from above by that of $(DS_k\text{-R})$ because all constraints of $(DS_k\text{-R})$ are present in (17) and (18), $\bar{y}_{jk} = 0$ for $j \notin J_k$, and $\sum_{t=H_k+1}^H \bar{v}_{kt} \leq 0$. This completes the proof, since the objective function value associated with $(\bar{\mathbf{u}}''_k, \bar{\mathbf{v}}''_k)$ in (DS_k) is clearly identical to that associated with the optimal solution $(\bar{\mathbf{u}}'_k, \bar{\mathbf{v}}'_k)$ in $(DS_k\text{-R})$. $\square$

The pseudocode of our Benders decomposition scheme with the cut strengthening feature for solving

(TR-A) is stated in Algorithms 1 and 2 in the online supplement, and Proposition 2 proves its correctness. The cut strengthening specified by the Steps 3–6 in Algorithm 2 has a pseudopolynomial time complexity of $\mathcal{O}(nH)$ with an overall complexity of $\mathcal{O}(mnH)$ for m machines. In practice, it is very fast.

In classical textbook applications of Benders decomposition, the current relaxed master problem is solved to optimality, and then cuts generated based on this optimal solution are added to it before the relaxed master problem is reoptimized. This loop is repeated until the optimality gap of (RMP)—the expression $(z(\bar{\mathbf{y}}) - \sum_{k=1}^m \bar{\eta}_k) / \sum_{k=1}^m \bar{\eta}_k$ —is smaller than a prespecified tolerance level, where the current optimal objective $\sum_{k=1}^m \bar{\eta}_k$ of (RMP) is a lower bound on that of (TR-A) and $z(\bar{\mathbf{y}})$ is the objective value of a feasible solution of (TR-A). The primary drawback of this classical scheme is that a new search tree is constructed every time the relaxed master problem is solved (Rubin 2011). Consequently, valuable time may be expended toward re-evaluating the same nodes over and over again. In contrast, using the *lazy constraint* technology offered by the state-of-the-art solvers allows us to execute the entire algorithm on a single search tree (IBM ILOG CPLEX 2011). In Step 11 of Algorithm 1, we invoke the *lazy constraint callback* routine for every candidate incumbent solution. The callback routine either identifies a missing Benders cut violated by the candidate solution and introduces it as a lazy constraint into the model or certifies the candidate as valid. Ultimately, no integer solution is evaluated multiple times during the course of the algorithm. Moreover, labeling the generated cuts as lazy informs the solver that most such constraints are not expected to be active at the optimal solution. Thus, we fully exploit the capabilities of the solver and allow it to apply the generated cuts as it deems necessary. The use of the lazy constraint technology appears to be relatively rare in the operations research literature, and we hope that it will be employed more frequently in the future, given that it may unleash the power of a cut generation algorithm, which seems impractical otherwise.

5. Computational Results

Outstanding among the accomplishments of this research is that both Rm-TWT, with a regular scheduling objective (see §5.1), and Rm-TWET with a non-regular scheduling objective (see §4.2 in the online supplement), are tackled successfully by the exact same solution approach. For both problems, the overarching goal of our computational study is to demonstrate that the proposed Benders-type method solves the preemptive relaxation (TR-A) to (near-) optimality in short computation times and provides tight lower bounds as well as high-quality job partitions for the original

problems. Very large instances of both problems are within the reach of our algorithm; however, we concede that the performance is somewhat better for Rm-TWT than for Rm-TWET.

The size of an instance is determined by the parameters m and n' so that the number of jobs is set to $n = mn'$. For each job $j \in \{1, \dots, n\}$, the processing time p_{j1} on the first machine is randomly drawn from the discrete uniform distribution $U[p_{\min}, p_{\max}]$. The processing times p_{jk} for $k \in \{2, \dots, m\}$ are then created as $\max(1, \lfloor U[1 - \theta, 1 + \theta] p_{j1} \rfloor)$. The earliness weight per unit time ϵ_j is generated from a discrete uniform distribution $U[\epsilon_{\min}, \epsilon_{\max}]$, and the corresponding unit tardiness weight is computed as $\lceil U[\alpha, \beta] \epsilon_j \rceil$. For Rm-TWT, all unit earliness weights are then set to zero. We generate the due dates by following a popular scheme in the literature (Liaw et al. 2003, Shim and Kim 2007a, Lin et al. 2011). The integral due date d_j of job j is calculated as $\lfloor U[\bar{P}(1 - TF - RDD/2)^+, \bar{P}(1 - TF + RDD/2)] \rfloor$, where the tardiness factor TF controls the tightness of the due dates and the due date range factor RDD determines their spread. The quantity $\bar{P} = (\sum_j \sum_k p_{jk})/m^2$ may be considered as the average load per machine. The parameters of the instance generation procedure are summarized in Table 1.

There are 12 combinations of the TF , RDD values and for each combination, five instances are generated. Therefore, we create 60 instances for each pair of m , n' values and a total of 720 instance pairs. The instances in a pair are identical, except that $\epsilon_j = 0$, $j = 1, \dots, n$ in the Rm-TWT instance. This data-generation scheme allows us to draw clear conclusions about the relative difficulty of Rm-TWET with respect to Rm-TWT. As pointed out by Kedad-Sidhoum et al. (2008), the motivation for the relatively large TF and small RDD values is that in most practical production environments the due dates are not loose and are not distant from each other. The rationale behind the selected $[\alpha, \beta]$ values reflects that the earliness cost is typically regarded as a finished goods inventory holding cost and should be less than the cost of loss of customer goodwill or a contractual penalty represented by the tardiness cost.

The computational results are obtained on a personal computer with a 3.80 GHz Intel Core i7 920 CPU with hyperthreading and 24 GB of memory running on Windows 7. Algorithms 1 and 2, which are collectively referred to as (TR-A)–BDS in this section, were implemented in C++ using the Concert Technology component library of IBM ILOG CPLEX 12.4. The cut generation procedure in Algorithm 2 is parallelized

through the Boost 1.51 library. More specifically, when a new integer-feasible solution is identified in the search tree for (RMP), m threads are constructed in the lazy constraint callback routine to solve (TR_k) , $k = 1, \dots, m$, in parallel. Note that in the presence of a control callback—such as the lazy constraint callback in (TR-A)–BDS—CPLEX applies a traditional branch-and-cut strategy by switching off its dynamic search feature and operates in an opportunistic parallel search mode. Following the termination of (TR-A)–BDS, we call the SiPS/SiPSi libraries (Tanaka et al. 2009, Tanaka and Fujikuma 2012) to solve m single-machine problems for each job partition in the final “solution pool” of CPLEX and obtain feasible solutions for Rm-TWT and Rm-TWET. Note that the current CPLEX engine generates and keeps multiple feasible solutions in addition to the optimal solution in a pool to help the user choose one that may fit criteria not represented explicitly in the current model solved (IBM ILOG CPLEX 2011). Furthermore, to promote the quality of the job partitions, the switch MIPemphasis in (TR-A)–BDS is set to 4 to urge CPLEX “to apply considerable additional effort toward finding high quality feasible solutions that are difficult to locate” (IBM ILOG CPLEX 2011). The source code of our algorithms and the test instances are either available from the corresponding author or at <http://people.sabanciuniv.edu/bulbul/publications.html>.

To justify the use of the proposed Benders-type approach to solve (TR-A), we benchmark it against (TR-A)–CPX, where the monolithic formulation (TR-A) is solved directly by invoking CPLEX. In this case, we let CPLEX decide whether to apply its dynamic search by running it with the default parameter settings, except that the opportunistic parallel search mode is turned on for a head-to-head comparison with (TR-A)–BDS. The relative gap tolerance parameter $EpGap$ of CPLEX is set to 3% while solving (TR-A) by either (TR-A)–CPX or (TR-A)–BDS. In addition, to illustrate the value of our approach in the absence of scalable alternate solution approaches for Rm-TWT and Rm-TWET in the literature and be able to provide more accurate optimality gaps for our lower and upper bounds, we also solve a time-indexed integer programming formulation for Rm-TWT and Rm-TWET via CPLEX under the default parameter settings. This formulation is referred to as (TI) in the following and obtained from that in Kedad-Sidhoum et al. (2008) by augmenting the time-indexed variables with a machine index and imposing a machine capacity constraint for each combination of machine and time period so that

Table 1 Instance-Generation Parameters

m	n'	$[p_{\min}, p_{\max}]$	θ	$[\epsilon_{\min}, \epsilon_{\max}]$	$[\alpha, \beta]$	TF	RDD
{2, 3, 4, 5}	{20, 30, 40}	[25, 100]	0.25	[1, 10]	[1.5, 3.0]	[0.4, 0.6, 0.8, 1.0]	[0.2, 0.4, 0.6]

no more than one job is in process at any instant on any machine. The best lower bound retrieved from (TI) at termination provides an alternate lower bound for the original nonpreemptive problems, and the best available objective value at termination provides a benchmark for the nonpreemptive solutions we construct for Rm-TWT and Rm-TWET.

All formulations are solved within the same working memory limit of 15 GB (WorkMem = 15,000). However, the memory footprint of (TR-A)–BDS does not exceed a few gigabytes, even for the largest instances with 200 jobs and five machines. The maximum number of threads that CPLEX is allowed to use—governed by the parameter Threads—is seven for all methods. The time limit parameter TiLim takes on the values 1,800, 1,800, and 600 seconds for (TI), (TR-A)–CPX, and (TR-A)–BDS, respectively.

Section 5.1 reports the results obtained for Rm-TWT. For ease of perusal, all tables employ a color formatting scheme so that the values of a performance indicator ranging from better to worse are indicated with shades of gray changing from light to dark (colors changing from green to red online).

5.1. Results for Rm-TWT

Table 2 consists of 12 parts, one for each possible combination of n and m listed in the first two columns. We report three types of percentage gaps in the table, labeled as “(TR-A)–BDS,” “LB quality,” and “Feasible solutions” in columns 4–12. The average times needed to solve the preemptive relaxation (TR-A) to within 3% of optimality by (TR-A)–CPX and (TR-A)–BDS are presented in columns 13–18. The color formatting is applied to these two sets of columns together to facilitate a head-to-head comparison. For each performance indicator, detailed results for each possible combination of TF and RDD values are included. The TF values appear in the third column, and the RDD values are specified in the column headers. All gaps larger than 100% are set to 100%, and the gap of a feasible solution with a positive objective function value with respect to a lower bound of 0 is assumed to be 100%. Each value in the table represents an average over five instances based on our data generation scheme, discussed previously.

The optimality gaps depicted in columns 4–6 are retrieved from CPLEX at the termination of (TR-A)–BDS, where CPLEX computes the optimality gap by taking the ratio of the best available lower bound to the objective value associated with the best integer solution at termination and then subtracting this ratio from 1. These results indicate that (TR-A)–BDS is able to solve the preemptive relaxation to the targeted precision of 3%. More specifically, (TR-A)–BDS terminates because of the time limit of 600 seconds for only 22 instances out of a total of 720. The corresponding number for (TR-A)–CPX is 47 with a time limit of 1,800 seconds.

The average (and median) gaps of (TR-A)–BDS and (TR-A)–CPX for those instances that could not be solved within the specified time limits are 7.22% (4.05%) and 74.14% (100%), respectively. Therefore, we conclude that the use of our Benders-type method for solving (TR-A) is well justified.

The next three columns under “LB quality” attest to the quality of the lower bound (LB) provided by (TR-A)–BDS for the optimal objective value of Rm-TWT. For a given instance, the expression (“Best Integer”–“LB”)/ (“Best Bound”) provides an upper bound on the gap of LB, where “Best Integer” and “Best Bound” are the objective function values associated with the best feasible solution available—retrieved from either our approach or (TI)—and the best lower bound provided by any one of the methods (TR-A)–CPX, (TR-A)–BDS, or (TI). For any n, m combination, the average LB gap summarized across all TF and RDD values does not exceed 8.15%, and the average LB gap across all instances is just 5.64%. In fact, only 8% of the instances (58 instances) has an LB gap larger than 10%.

The following three columns under “Feasible solutions” present the average upper bounds on the optimality gaps attained by our nonpreemptive feasible solutions. For a given feasible solution, an upper bound on the optimality gap is calculated as (“0FV”–“Best Bound”)/ (“Best Bound”), where “0FV” is the objective function value of the feasible solution. We do not include detailed results about (TI) but note that the incumbent from (TI) is hardly competitive with the best feasible solution obtained from (TR-A)–BDS, except for the 40-job instances. Moreover, even the LP relaxation of (TI) is not solved within a half hour for instances with 100 or more jobs. The average (median) optimality gaps over all instances solved are 3.55% (1.73%) and 30.28% (10.20%) for (TR-A)–BDS and (TI), respectively. Perhaps more importantly, the proposed approach delivers a robust performance and scales to very large instances. With the exception of a little over 4% of the instances (31 of 720), the optimality gap is always below 10%. The corresponding number for (TI) is 50% (181 of 360).

The relatively higher gaps under “LB quality” and “Feasible solutions” in Table 2 for TF = 0.4 stem from the small objective function values associated with loose due dates. Even small errors result in large percentage gaps in this case. Note that the objective function value of an instance with TF = 0.6, 0.8, and 1.0 is on average 7.5, 25.1, and 45.9 times larger, respectively, compared to that of an instance with TF = 0.4. A second contributing factor here is the growing size of (TI) with looser due dates. Frequently, even the LP relaxation is not solved within the allotted time for such instances, and this results in smaller “Best Bound” values in general. In other words, the actual performance for TF = 0.4 is probably better than what it appears to be.

Table 2 (Color online) Results for Rm-TWT

$n \quad m \quad \text{TF}$			Percentage gaps									(TR-A)–Solution times					
			(TR-A)–BDS			LB quality			Feasible solutions			CPX			BDS		
			0.2	0.4	0.6	0.2	0.4	0.6	0.2	0.4	0.6	0.2	0.4	0.6	0.2	0.4	0.6
40	2	0.4	1.3	1.6	0.5	3.0	7.5	13.2	1.6	4.5	6.3	2	3	10	2	2	3
		0.6	2.0	2.2	2.2	2.8	3.5	6.4	1.0	1.8	2.2	4	3	4	2	2	3
		0.8	2.5	2.6	2.4	2.8	3.6	3.7	0.8	0.6	0.6	6	5	5	1	1	3
		1.0	2.4	2.1	2.2	2.3	2.4	2.3	0.5	0.5	0.8	6	6	5	1	1	1
60	2	0.4	2.2	1.8	1.6	3.5	5.3	9.5	0.9	2.0	5.0	8	7	11	5	7	10
		0.6	2.4	2.4	2.6	3.3	4.4	5.7	0.9	1.3	1.7	15	12	11	5	7	18
		0.8	2.7	1.7	2.5	3.1	2.7	4.4	1.2	0.6	1.2	21	19	17	3	6	9
		1.0	2.2	2.8	2.8	1.9	2.2	2.5	0.6	1.2	1.0	24	25	24	2	2	2
	3	0.4	2.4	2.1	1.4	5.0	9.3	42.9	1.9	3.6	36.8	20	60	215	2	4	4
		0.6	2.7	2.7	2.8	3.8	5.0	8.7	1.5	1.7	3.0	32	30	36	1	2	12
		0.8	2.8	2.8	2.6	3.1	4.3	4.8	1.4	0.9	0.9	40	40	36	1	2	3
		1.0	2.6	2.5	2.5	2.2	2.6	3.2	0.8	0.6	0.8	41	44	41	1	1	1
	4	0.4	2.0	2.2	2.0	3.0	5.0	11.6	1.1	2.0	6.8	15	12	44	8	17	20
		0.6	2.2	1.4	2.6	2.9	2.8	4.2	1.0	1.1	0.9	33	27	30	10	21	36
		0.8	2.9	2.3	2.7	3.3	3.0	4.3	1.5	0.7	1.1	52	56	44	3	9	10
		1.0	2.2	2.4	2.3	1.6	2.7	2.7	1.1	0.4	0.6	69	63	59	3	6	7
80	4	0.4	2.7	2.6	1.2	6.8	10.4	38.0	3.7	5.0	36.6	42	295	1716	5	7	11
		0.6	2.7	2.5	3.8	4.2	5.5	10.1	1.6	2.1	4.2	79	73	484	2	7	462
		0.8	2.8	2.4	2.9	3.6	4.3	5.9	1.1	0.7	1.0	111	104	98	1	6	11
		1.0	2.5	2.8	2.8	2.2	3.1	3.7	0.7	1.1	0.7	118	112	121	1	1	2
	3	0.4	2.2	2.2	0.9	3.8	7.3	26.9	1.6	3.5	21.2	34	112	1030	5	11	28
		0.6	2.6	2.8	2.9	3.3	4.3	6.3	1.2	1.4	2.1	93	87	93	2	5	24
		0.8	2.6	2.5	2.6	3.3	3.8	4.5	1.1	0.9	1.2	122	121	117	2	3	5
		1.0	2.6	2.4	2.4	2.1	3.0	3.2	1.2	0.8	1.0	148	142	138	2	2	2
	5	0.4	2.7	2.7	2.8	7.3	13.7	20.0	6.2	13.3	20.0	104	1119	1800	6	13	125
		0.6	2.7	2.8	5.9	4.2	5.7	12.8	2.9	4.6	9.3	149	168	883	4	15	601
		0.8	2.6	2.8	3.1	3.7	4.9	6.5	2.3	3.5	4.9	235	182	242	2	15	308
		1.0	2.5	2.5	2.8	2.8	3.3	4.3	1.4	1.9	2.7	253	233	235	2	2	3
120	3	0.4	2.5	1.9	1.1	3.7	6.5	42.5	2.3	5.8	42.5	48	66	1367	7	20	109
		0.6	2.2	2.6	2.4	2.9	4.2	5.6	1.6	2.9	4.6	205	172	162	4	8	49
		0.8	2.6	2.9	2.7	3.0	4.0	4.5	1.5	2.3	2.8	262	254	221	4	5	8
		1.0	2.8	2.3	2.6	3.0	2.7	3.3	1.1	1.3	1.4	338	344	303	3	4	6
	4	0.4	2.2	2.9	10.8	4.1	9.1	20.0	3.0	7.9	20.0	96	516	1816	13	18	135
		0.6	2.7	2.6	3.1	3.6	4.4	7.4	2.4	2.8	5.9	205	209	327	4	18	225
		0.8	2.5	2.5	2.9	3.2	4.1	5.0	1.9	2.6	3.2	316	315	287	3	6	16
		1.0	2.7	2.7	2.7	3.0	3.2	3.7	1.4	1.6	2.3	346	326	291	2	3	3
	5	0.4	2.5	2.9	0.0	5.2	10.0	0.0	4.0	9.0	0.0	216	1356	1527	19	35	18
		0.6	2.7	2.7	3.8	3.7	4.6	8.8	2.1	3.3	6.6	380	379	804	5	39	415
		0.8	2.6	2.7	2.9	3.3	4.3	5.5	1.8	2.8	4.1	683	646	600	4	11	48
		1.0	1.3	2.5	2.4	1.6	3.1	3.5	0.8	1.7	2.3	752	713	653	5	4	4
160	4	0.4	2.8	2.8	0.0	4.5	9.2	0.0	3.1	7.9	0.0	143	451	1361	9	40	29
		0.6	2.7	2.9	2.9	3.3	4.2	6.4	1.6	2.7	4.8	429	302	704	6	11	176
		0.8	2.6	2.2	2.6	3.1	3.3	4.3	1.6	2.2	2.8	713	742	668	5	9	12
		1.0	2.0	2.0	2.6	2.2	2.4	3.3	0.9	1.1	1.8	934	836	851	5	6	6
	5	0.4	2.5	3.4	0.0	4.5	12.3	0.0	3.3	10.8	0.0	345	1476	793	24	324	35
		0.6	2.5	2.8	3.9	3.2	4.5	7.9	1.7	2.7	5.4	964	752	1327	11	33	510
		0.8	2.6	2.7	2.7	3.1	3.9	4.6	1.9	2.2	3.0	1720	1708	1440	7	29	36
		1.0	2.4	2.0	2.3	2.7	2.4	3.1	2.5	1.8	2.1	1803	1771	1719	7	9	9
200	5	0.4	2.5	3.4	0.0	4.5	12.3	0.0	3.3	10.8	0.0	345	1476	793	24	324	35
		0.6	2.5	2.8	3.9	3.2	4.5	7.9	1.7	2.7	5.4	964	752	1327	11	33	510
		0.8	2.6	2.7	2.7	3.1	3.9	4.6	1.9	2.2	3.0	1720	1708	1440	7	29	36
		1.0	2.4	2.0	2.3	2.7	2.4	3.1	2.5	1.8	2.1	1803	1771	1719	7	9	9

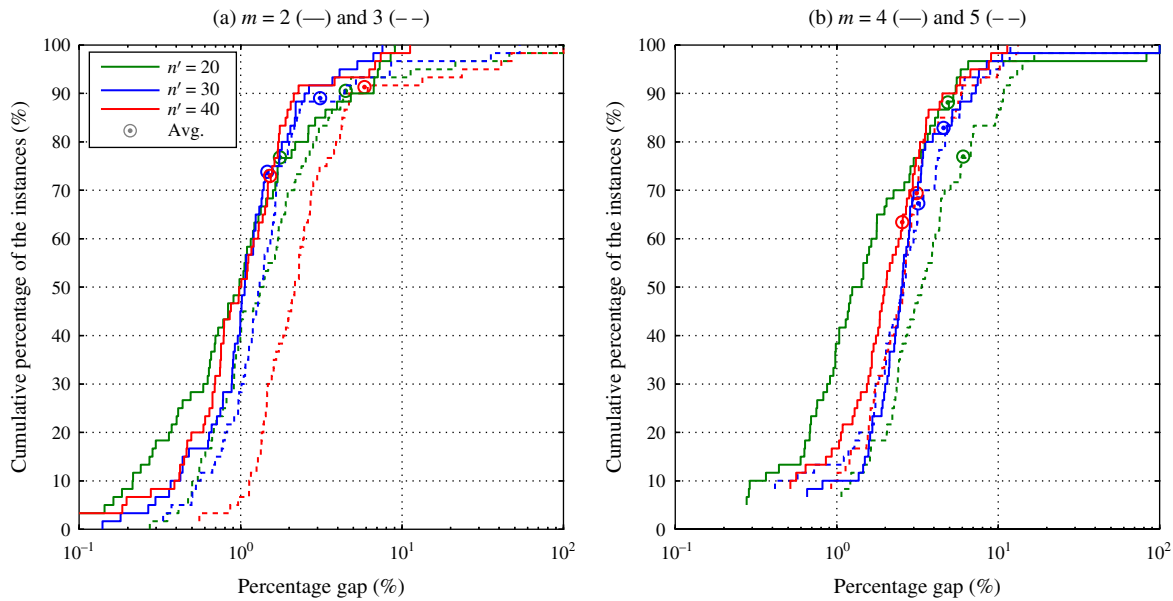


Figure 1 (Color online) The Empirical Distributions of the Optimality Gaps of the Upper Bounds by (TR-A)-BDS for Rm-TWT

The robustness of the quality of the feasible solutions obtained from our Benders-type approach is further illustrated in Figures 1(a)–1(b). The empirical distributions of the optimality gaps of the feasible solutions associated with (TR-A)-BDS are plotted in these figures. The horizontal axes are in logarithmic scale to increase the readability of the graph. Note that the median percentage gap for each curve corresponds to the 50% mark on the vertical axis, and the average gaps are explicitly indicated. The curves are clustered and rise steeply. That is, the quality of the partitions retrieved from (TR-A)-BDS is not particularly sensitive to the increasing number of jobs n' per machine.

The solution time performance of (TR-A)-BDS is overwhelmingly superior to that of (TR-A)-CPX. Based on the instances that are solved by both methods within the time limit, the ratio of the solution time of (TR-A)-CPX to that of (TR-A)-BDS is 46.7 on average. From a total of 720 instances, only 35 take slightly more time to solve for (TR-A)-BDS than for (TR-A)-CPX. For both methods, instances with loose average due dates within a relatively wide range are more problematic. However, tightening the due dates also hurts the performance of (TR-A)-CPX, whereas it benefits that of (TR-A)-BDS. A detailed analysis of the empirical distributions of the solution times of (TR-A)-BDS and (TR-A)-CPX is available in §4.1 of the online supplement.

Recall that we call the SiPS/SiPSi libraries (Tanaka et al. 2009, Tanaka and Fujikuma 2012) to solve m single-machine problems for each job partition present in the final solution pool of CPLEX and obtain feasible solutions for Rm-TWT and Rm-TWET following the termination of (TR-A)-BDS. We do not report detailed results for the sake of brevity. However, our use of an

optimal algorithm to solve the single-machine problems for a given job partition is well justified. Even for the five machine and 200 job instances, it takes an average of 2.31 seconds and no more than 6.96 seconds to solve all single-machine problems to optimality by the SiPS/SiPSi solver for all job partitions identified. This solver is extremely fast; the time expended for a 40-job single machine instance is about 30 milliseconds. We emphasize that the best solution of the preemptive relaxation does not necessarily produce the best nonpreemptive solution for the original problem. Therefore, the ability of locating many high-quality job partitions in the search tree is a critical advantage of (TR-A)-BDS, which identifies on average 4.7 times more job partitions per instance than (TR-A)-CPX. This characteristic may also prove useful to jump start a population-based heuristic following the completion of (TR-A)-BDS. In summary, coupled with its demonstrated ability to construct high-quality lower and upper bounds for the original problem, this approach's outstanding total solution time performance makes it a viable alternative for tackling very large instances of Rm-TWT successfully.

We conclude this section with a brief discussion on the time-indexed formulation (TI). The main purpose of solving (TI) in this paper is to incorporate the objective function value of the incumbent solution and the best lower bound available at termination into the “Best Integer” and “Best Bound” values, respectively, so that we quantify the gaps of our lower and upper bounds as accurately as possible. Otherwise, solving (TI) is not a scalable solution approach for Rm-TWT, as discussed previously. In addition, the LP relaxation of (TI) also suffers from the same scalability issue as a lower bounding method. We provide further specifics and settle this issue here.

On the one hand, the LP relaxations of time-indexed formulations are strong and provide very tight bounds. On the other hand, the size of a time-indexed formulation grows with the length of the planning horizon and is therefore pseudopolynomial. From a computational perspective, the solution effort expended increases rapidly with longer processing times, and CPLEX cannot solve the LP relaxation of (TI) or find any feasible solution within the time limit of 1,800 seconds for the Rm-TWT (and Rm-TWET) instances with more than 90 jobs. For the sake of completeness, we benchmarked the lower bound produced by (TR-A)–BDS against the optimal objective function value of the LP relaxation of (TI). These two lower bounds do not dominate each other. There are instances in which the objective value of the LP relaxation of (TI) is larger than that of (TR-A), and vice versa. The best lower bound retrieved from (TR-A)–BDS at termination is on average 97.04% of the optimal objective value of the LP relaxation of (TI), computed over 360 Rm-TWT instances with $n \leq 90$. The corresponding figure for the Rm-TWET instances is 94.29%. Furthermore, recall that (TR-A)–BDS terminates with a 3% relative optimality gap. Therefore, it is fair to state that the lower bounds provided by (TR-A)–BDS and the LP relaxation of (TI) are of comparable quality. Ultimately, (TR-A)–BDS is the clear choice as a lower bounding technique, given its superior computational time performance and the high quality of the nonpreemptive schedules based on the solution of (TR-A)–BDS.

5.2. Results for Rm-TWET

The results for Rm-TWET are relegated to §4.2 of the online supplement. For ease of perusal, all tables employ a color formatting scheme so that the values of a performance indicator ranging from better to worse are indicated with shades of gray changing from light to dark (colors changing from green to red online).

6. Conclusions and Future Research

In this paper, we developed a new preemptive relaxation for unrelated parallel machine scheduling problems with weighted tardiness and weighted earliness/tardiness objectives. The key property of this relaxation is that it provides a tight lower bound and a set of high-quality job partitions that forms the basis for the near-optimal nonpreemptive solutions for the original problem. The relaxation itself is formulated as a difficult mixed-integer linear program, and a computationally effective Benders decomposition algorithm that can handle very large instances of this formulation is a primary contribution of this paper. Our implementation employs state-of-the-art computational features, such as the *lazy constraint* callback of IBM ILOG CPLEX (2011) and a parallelization of the Benders subproblems

via the Boost 1.51 library. Ultimately, we characterize our approach as a simple, nonparametric, and easy-to-implement mathematical programming-based heuristic with a further distinguishing property that it can handle both a regular and nonregular scheduling objective successfully with no additional customization. The results for Rm-TWT are outstanding. Although those for Rm-TWET are not on a par, we determine that they are of high quality.

Initially, we also experimented with the identical parallel machine scheduling problems $Pm // \sum_j \pi_j T_j$ and $Pm // \sum_j \pi_j T_j + \epsilon_j E_j$. However, the symmetry inherent in these problems results in many similar cuts and causes (TR-A)–BDS to choke. One of the items in our future research agenda is to explore ways of enhancing our algorithm so it can handle the identical parallel machine environment.

A further goal is to embed (TR-A)–BDS into an optimal algorithm for Rm-TWT and Rm-TWET. Note that the proposed preemptive relaxation can naturally handle branching decisions on the job to machine assignments.

Supplemental Material

Supplemental material to this paper is available at <http://dx.doi.org/10.1287/ijoc.2014.0615>.

Acknowledgments

The authors are grateful for the invaluable comments of the anonymous referees that helped to improve the paper.

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